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Reimbursed use of GLP-1 receptor agonists in France, 2019–2025

National, active-substance, demographic and regional trends from Open Medic data

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An open, reproducible drug-utilisation study of reimbursed community dispensings in France.

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Document status

This document reports a descriptive analysis conducted under a protocol and statistical analysis plan frozen before substantive results were examined. Three documented amendments address a historical Anatomical Therapeutic Chemical (ATC) coding discontinuity, figure-linked data and extended regional outputs, and reproducible geospatial presentation. The analysis is intended for transparent scientific communication and hypothesis generation; it is not a regulatory assessment, clinical guideline or causal study.

Key findings

- Annual beneficiaries with at least one reimbursed GLP-1 receptor agonist dispensing increased from 411,769 in 2020 to 925,644 in 2025 (+124.8%).
- Growth continued across the study period, with a temporary slowdown in 2024 before renewed expansion in 2025.
- Semaglutide rose from 4.8% of reimbursed boxes in 2019 to 53.0% in 2025, while liraglutide and exenatide declined markedly.
- Rates remained highest among people aged 60 years or older, particularly men, although relative growth occurred in every age-sex group.
- In 2025, metropolitan age-sex-standardised rates varied twofold, from 868.2 per 100,000 in Bretagne to 1,747.2 in Hauts-de-France.
- The combined overseas grouping had a standardised rate of 2,435.9 per 100,000. Dedicated quality control found no known computational or file-version explanation, but the aggregate cannot be assigned to individual territories and warrants cautious interpretation.

Contents

Executive summary

Background: Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) are increasingly important in type 2 diabetes and weight management. Their rapid uptake, high acquisition costs, potential population-level budget impact and recurrent supply constraints make timely utilisation surveillance a public-health and health-system priority.

Methods: We conducted a repeated annual cross-sectional study of aggregated Open Medic reimbursement data for community-pharmacy dispensings in France. The primary pharmacological definition was ATC level 4 code A10BJ. National beneficiary counts and rates were analysed for 2020–2025; reimbursed boxes and nominal expenditure were analysed for 2019–2025 after historical ATC-code harmonisation. Rates used average annual population estimates from INSEE. Regional rates were directly standardised by age and sex to the 2025 population of France.

Results: Annual beneficiaries increased from 411,769 (609.4 per 100,000 residents) in 2020 to 925,644 (1,342.2 per 100,000) in 2025, corresponding to increases of 124.8% and 120.2%, respectively. Reimbursed boxes increased from 3.49 million in 2019 to 8.83 million in 2025, and nominal reimbursed expenditure from €298.7 million to €674.6 million. Semaglutide accounted for 53.0% of boxes and 51.6% of expenditure in 2025. Rates were highest among adults aged 60 years or older. Metropolitan 2025 standardised rates ranged from 868.2 to 1,747.2 per 100,000. The combined overseas grouping reached 2,435.9 per 100,000 but could not be disaggregated geographically. Growth slowed temporarily in 2024 before resuming in 2025.

Interpretation: Reimbursed GLP-1 RA use more than doubled in France over five years, alongside a major shift toward semaglutide and persistent demographic and regional heterogeneity. The temporary 2024 slowdown coincided with documented supply constraints, although aggregate annual data cannot establish causation. Continued surveillance should distinguish reimbursed diabetes-oriented use from non-reimbursed obesity treatment, incorporate newer incretin therapies and improve territorial resolution for overseas France.

Introduction

Glucagon-like peptide-1 receptor agonists have moved from a relatively specialised glucose-lowering class to a central component of contemporary metabolic medicine. In addition to improving glycaemic control, newer agents produce clinically meaningful weight loss and, for semaglutide in selected patients with overweight or obesity and established cardiovascular disease, reduce major cardiovascular events [1,2]. These benefits have broadened clinical interest well beyond diabetes care and driven rapid growth in demand.

Real-world utilisation has risen across health systems. Scandinavian studies documented rapid uptake of semaglutide and changing user profiles between 2018 and 2023 [3,4]. In France, du Soulier and colleagues used individual-level national health data to show that incident GLP-1 RA use rose substantially from 2017 through mid-2023 and that a small but increasing fraction of users had neither recorded type 2 diabetes nor obesity-related conditions [5]. Epi-Phare separately described 7,048 adults treated with Wegovy during its early-access period, most of whom were women and had severe obesity with weight-related comorbidity; initiation also varied geographically [6]. Building on this evidence, we extend the descriptive assessment through the full 2025 calendar year and provide a fully open and reproducible view of national, demographic, active-substance and regional reimbursement trends in France.

The scale of use has direct economic relevance. GLP-1 RAs are comparatively costly therapies intended for chronic use, and rapid diffusion can materially affect pharmaceutical expenditure even when targeted to high-risk groups. The Haute Autorité de santé judged the three-year budget impact of reimbursing semaglutide for a restricted severe-obesity population to be highly uncertain; under its alternative population estimate, the impact was approximately €2.8 billion [7]. This drug-budget question sits within the wider economic burden of overweight and obesity, which the OECD has estimated to account for approximately 8% of health expenditure and 3.3% of gross domestic product across member countries [8]. Although these estimates measure different economic dimensions and are not directly comparable, they collectively underscore the importance of monitoring treated populations, dispensed volumes and reimbursed expenditure. Tracking these complementary indicators is necessary to inform equitable access, anticipate supply requirements and assess the longer-term financial implications for the health system.

Surveillance is also important because exceptional demand has strained supply. French and European regulators reported recurrent shortages from 2022 onward; France suspended or discouraged new initiations of several agents during parts of late 2023 and 2024 to protect continuity of care [9,10]. The same period was marked by concern about non-indicated or cosmetic use, although reimbursement data alone incompletely capture that phenomenon [11]. Annual utilisation patterns must therefore be interpreted in the context of product availability, regulatory measures, changing indications and evolving public awareness.

Open Medic, published by the French National Health Insurance, provides aggregated annual data on reimbursed medicines dispensed by community pharmacies [12]. Linked to population denominators from INSEE, it allows transparent estimation of beneficiary rates and regional variation without access to individual-level health records. We used these open data to describe reimbursed GLP-1 RA use in France through 2025, nationally, by active substance, by age and sex, and across regional analytical groupings. We also examined temporal variation in growth and the interpretive limitations created by the combined reporting of overseas regions and departments.

Methods

Study design and governance

This was a repeated annual cross-sectional drug-utilisation study using aggregated administrative reimbursement data. Each calendar year was treated as a separate observation period; individuals could not be linked across years, substances or geographic units. The protocol and statistical analysis plan were frozen on 27 August 2026 before substantive results were examined. Analyses were subsequently governed by three version-controlled amendments. No null-hypothesis tests were planned because the objective was descriptive and the source comprised administrative counts rather than a probability sample.

Data sources

Open Medic complementary databases supplied annual beneficiary counts, reimbursed boxes, reimbursement base and reimbursed expenditure for medicines dispensed in community pharmacies and reimbursed by mandatory French health-insurance schemes. ATC level 4 files supported class-level analyses and ATC level 5 files supported active-substance analyses. Source archives, download dates, URLs, file sizes and SHA-256 checksums were recorded in a machine-readable manifest. Corrected regional files for 2023–2025 were verified, including the 2025 release corrected after 10 July 2026 [12].

Population denominators came from INSEE regional estimates by sex and five-year age group [13]. The average population for year y was calculated as the mean of population estimates on 1 January of year y and 1 January of year y+1. Geographic boundaries used in maps came from the 2025 Etalab Contours administratifs dataset at 100-metre generalisation, derived from IGN ADMIN EXPRESS [14].

Study population and pharmacological definition

The numerator population comprised beneficiaries with at least one reimbursed community dispensing during the calendar year and at the relevant aggregation level. The denominator comprised residents in the corresponding INSEE population estimate. Rates are therefore population-based rates of reimbursed-dispensing beneficiaries, not disease prevalence or treatment prevalence in a closed cohort.

The primary definition was ATC level 4 code A10BJ (glucagon-like peptide-1 analogues). The substance analysis included exenatide (A10BJ01), liraglutide (A10BJ02), dulaglutide (A10BJ05) and semaglutide (A10BJ06), the substances observed during the study period. Tirzepatide (A10BX16), a dual glucose-dependent insulinotropic polypeptide/GLP-1 receptor agonist, and fixed insulin–GLP-1 combinations were excluded. Substance-specific beneficiary counts were not summed because one person could receive more than one substance in a year.

Historical ATC harmonisation and analytical windows

Quality control identified a historical classification discontinuity in 2019: the ATC4 A10BJ record contained semaglutide only, while other GLP-1 RAs remained coded under historical A10BX categories. Protocol amendment 001 therefore reconstructed additive 2019 measures (boxes and expenditure) from harmonised ATC5 records and treated the unique class-level beneficiary count as not estimable. Consequently, national, demographic and regional beneficiary analyses cover 2020–2025, whereas additive substance, box and expenditure analyses cover 2019–2025.

Measures and statistical analysis

The primary measure was the annual number of class-level beneficiaries and the rate per 100,000 average residents. Secondary measures included reimbursed boxes, nominal reimbursed expenditure, year-on-year changes, endpoint percentage changes and compound annual growth rates. Boxes were treated as dispensing-volume units, not equivalent treatment quantities, because package sizes, doses and dosing schedules differ across products and time.

Age- and sex-specific rates were calculated for six prespecified strata: females and males aged 0–19 years, 20–59 years, and 60 years or older. Unknown age and sex categories were retained for quality control but excluded from denominator-based analyses without redistribution or imputation.

Regional analyses used 13 Open Medic analytical units: 12 metropolitan groupings and one combined overseas grouping. Provence-Alpes-Côte d'Azur and Corse form one Open Medic unit. Crude rates used directly reported regional beneficiaries. Directly age- and sex-standardised rates applied six national 2025 population weights to stratum-specific regional rates. Where disclosure control made a young age-sex cell ambiguous in 2020–2021, conservative bounds were produced by allocating between zero and all compatible unknown-age beneficiaries to that cell; these are data-uncertainty intervals, not confidence intervals. Maps were restricted to 2022–2025, when complete point estimates were available.

Beneficiaries were independently deduplicated within each annual aggregation level. Consequently, regional beneficiary counts are not additive: a person dispensed medicines in more than one regional grouping can contribute once to each relevant regional count but only once to the national count. Additive measures such as boxes and reimbursed expenditure reconcile across geographic levels. This distinction was retained in all regional comparisons and reconciliation checks.

Overseas grouping audit

Because the combined overseas estimate was substantially higher than metropolitan values, a dedicated audit independently checked numerator reconciliation, denominator construction and standardisation. INSEE denominators for Guadeloupe, Martinique, Guyane, La Réunion and Mayotte were summed and compared with the direct overseas aggregate for every year-age-sex cell. Direct and stratified Open Medic numerators, crude rates and standardised rates were reproduced independently. Source-file checksums and the timing of corrected releases were verified. No territory-specific rate was inferred from the combined result.

Reproducibility, reporting and ethics

Every figure was generated from a figure-specific machine-readable CSV and registered in a manifest. Analytical scripts used R and Python with locked dependencies; source provenance and quality-control outputs were version-controlled. Changes were calculated from unrounded values; displayed components may therefore not sum exactly to totals, and displayed differences may not equal differences calculated from rounded values. This report was prepared with reference to the RECORD-PE reporting guideline [18]. A completed checklist, including items considered not applicable to this aggregate-data study, is provided in the supplementary materials.

The study used only open, aggregated, disclosure-controlled data and involved no identifiable individual-level records. Research ethics approval and individual consent were therefore not required. The analytical repository, including code, metadata, quality-control outputs and figure-linked data, is intended for public release at https://github.com/javierzsm/glp1-use-france.

Results

National evolution

The number of annual beneficiaries increased in every study year, from 411,769 in 2020 to 925,644 in 2025 (Table 1; Figure 1). This was an absolute increase of 513,875 beneficiaries and a relative increase of 124.8%, equivalent to a compound annual growth rate of 17.6%. The population rate rose from 609.4 to 1,342.2 per 100,000 residents (+120.2%). Growth was strong from 2020 through 2023 (annual beneficiary increases of 20.2%–23.7%), slowed to 3.6% in 2024, and reaccelerated to 18.9% in 2025.

Table 1. Annual beneficiaries and population rates for reimbursed GLP-1 receptor agonist dispensings in France, 2020–2025.

Year	Beneficiaries	Rate per 100,000	Annual change, count	Annual change, rate
2020	411,769	609.4	—	—
2021	509,236	750.2	23.7%	23.1%
2022	624,721	915.6	22.7%	22.0%
2023	751,016	1,096.0	20.2%	19.7%
2024	778,285	1,132.1	3.6%	3.3%
2025	925,644	1,342.2	18.9%	18.6%

Note: Beneficiaries had at least one reimbursed community dispensing during the calendar year. Rates use the estimated average annual population. Sources: Open Medic; INSEE.

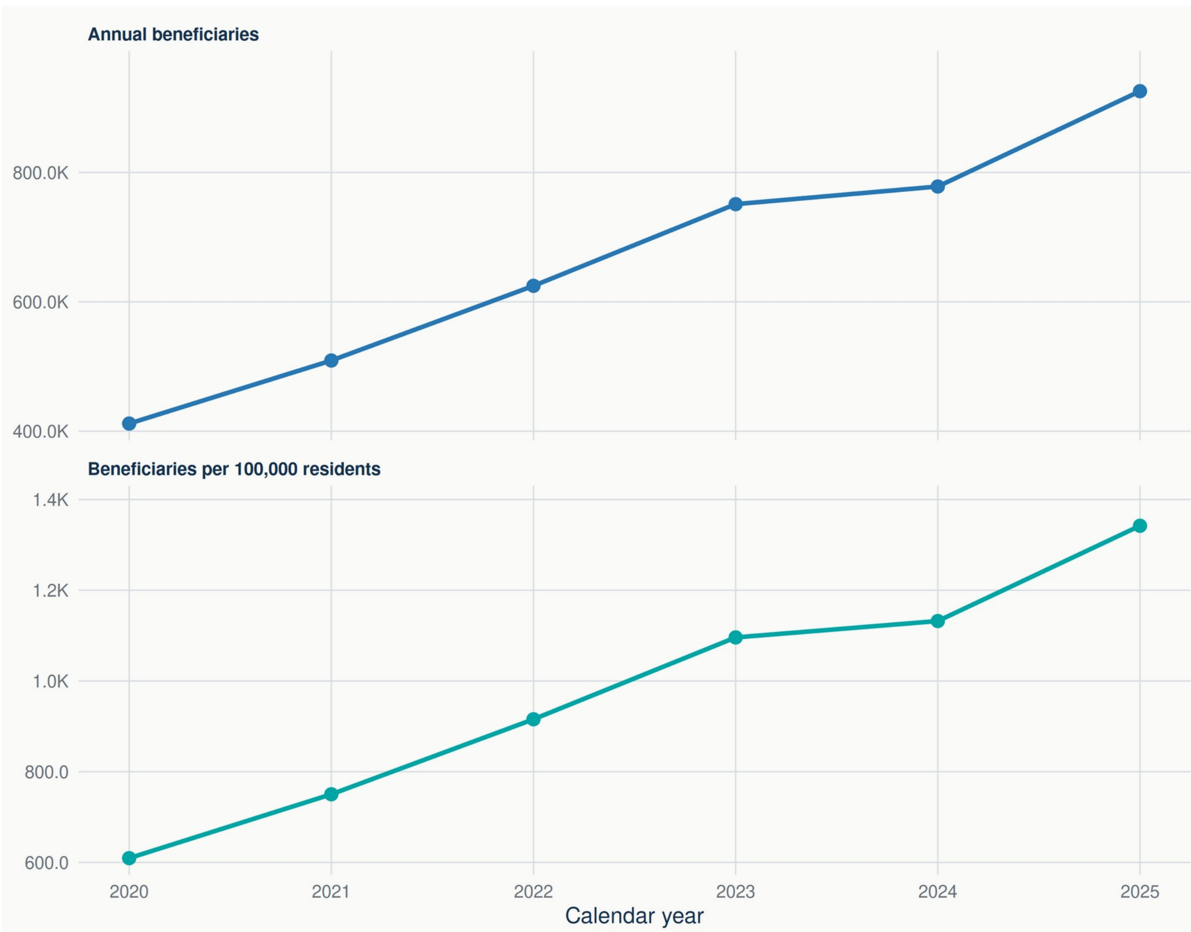

Figure 1. Annual beneficiaries and beneficiary rates for reimbursed GLP-1 receptor agonist dispensings in France, 2020–2025. Panels use independent vertical scales. Beneficiaries had at least one reimbursed community dispensing during the calendar year. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Active-substance volumes and expenditure

Reimbursed boxes increased from 3.49 million in 2019 to 8.83 million in 2025 (+152.9%; compound annual growth rate 16.7%), while nominal reimbursed expenditure increased from €298.7 million to €674.6 million (+125.9%; compound annual growth rate 14.5%). The class composition changed substantially (Table 2; Figure 2). Semaglutide increased from 168,797 boxes (4.8%) in 2019 to 4.68 million (53.0%) in 2025. Dulaglutide increased in absolute terms from 1.53 million to 3.76 million boxes but its share was 42.6% in 2025. Liraglutide declined from 47.4% to 4.4% of boxes, while exenatide became negligible.

The national box total was almost unchanged between 2023 and 2024 (+0.1%). This net plateau concealed opposing substance-level changes: semaglutide added 719,852 boxes, while liraglutide and dulaglutide declined by 424,938 and 285,488 boxes, respectively. In 2025, semaglutide added a further 1.04 million boxes and dulaglutide rebounded by 356,461 boxes, producing an overall increase of 18.0%. Nominal reimbursed expenditure declined by 1.2% in 2024 before increasing by 15.9% in 2025.

Table 2. Active-substance composition of reimbursed GLP-1 receptor agonist boxes and nominal expenditure, France, 2019 and 2025, with contributions to recent annual box changes.

Substance	2019 boxes (share)	2025 boxes (share)	Δ boxes 2023–24	Δ boxes 2024–25	2025 reimbursed expenditure
Exenatide	140,950 (4.0%)	714 (0.0%)	-2,090	-11,195	€0.1m
Liraglutide	1,654,371 (47.4%)	392,753 (4.4%)	-424,938	-35,758	€26.2m
Dulaglutide	1,526,933 (43.7%)	3,758,813 (42.6%)	-285,488	+356,461	€300.3m
Semaglutide	168,797 (4.8%)	4,677,641 (53.0%)	+719,852	+1,035,195	€348.1m

Note: Historical 2019 substance codes were harmonised under protocol amendment 001. Box counts are additive; substance-specific beneficiary counts are not. Expenditure is nominal. Displayed components may not sum exactly to totals because values are rounded independently. Source: Open Medic.

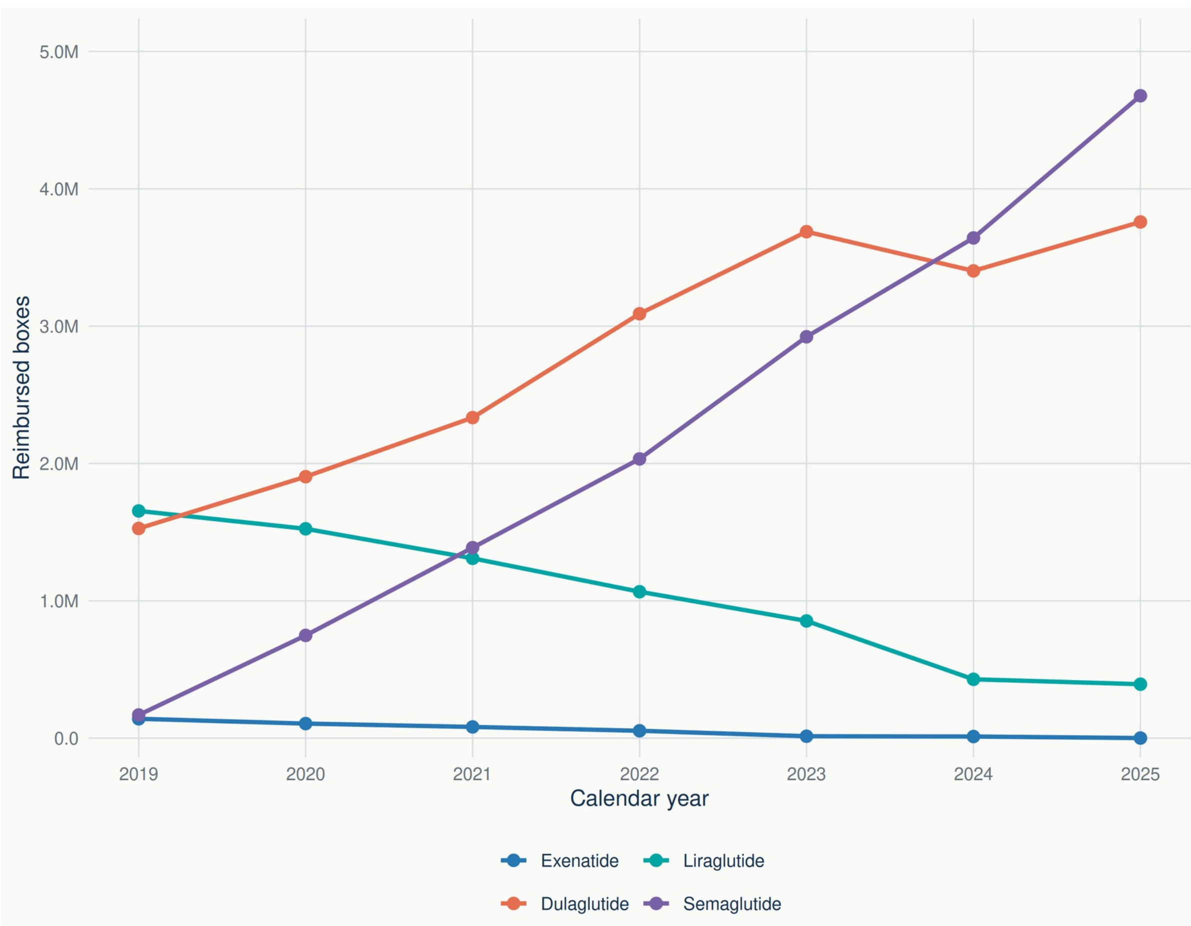

Figure 2. Annual reimbursed boxes of GLP-1 receptor agonists by active substance in France, 2019–2025. Historical 2019 substance codes were harmonised according to protocol amendment 001. Sources: Open Medic.

Age and sex

Rates increased in all six age-sex groups (Table 3; Figure 3). In 2025, the highest rate was among males aged 60 years or older (3,719.5 per 100,000), followed by females in the same age group (2,459.0). Among people aged 20–59 years, rates were similar in 2025 for males (993.1) and females (974.8), although the relative increase since 2020 was greater among females (+129.5% versus +102.9%). Rates among people aged 0–19 years remained very low—8.4 for males and 13.5 for females—so their large relative increases should not be interpreted as a large absolute burden.

Table 3. Change in age- and sex-specific beneficiary rates for reimbursed GLP-1 receptor agonist dispensings, France, 2020–2025.

Age group	Sex	2020 rate	2025 rate	Absolute change	Relative change
0-19 years	Male	2.1	8.4	6.3	302.9%
0-19 years	Female	3.2	13.5	10.2	315.2%
20-59 years	Male	489.4	993.1	503.8	102.9%
20-59 years	Female	424.8	974.8	550.0	129.5%
60 years or older	Male	1,767.5	3,719.5	1,952.0	110.4%
60 years or older	Female	1,173.5	2,459.0	1,285.5	109.5%

Note: Rates are per 100,000 residents in the corresponding age-sex stratum. Changes were calculated from unrounded values and may differ slightly from subtraction of the displayed rates. Sources: Open Medic; INSEE.

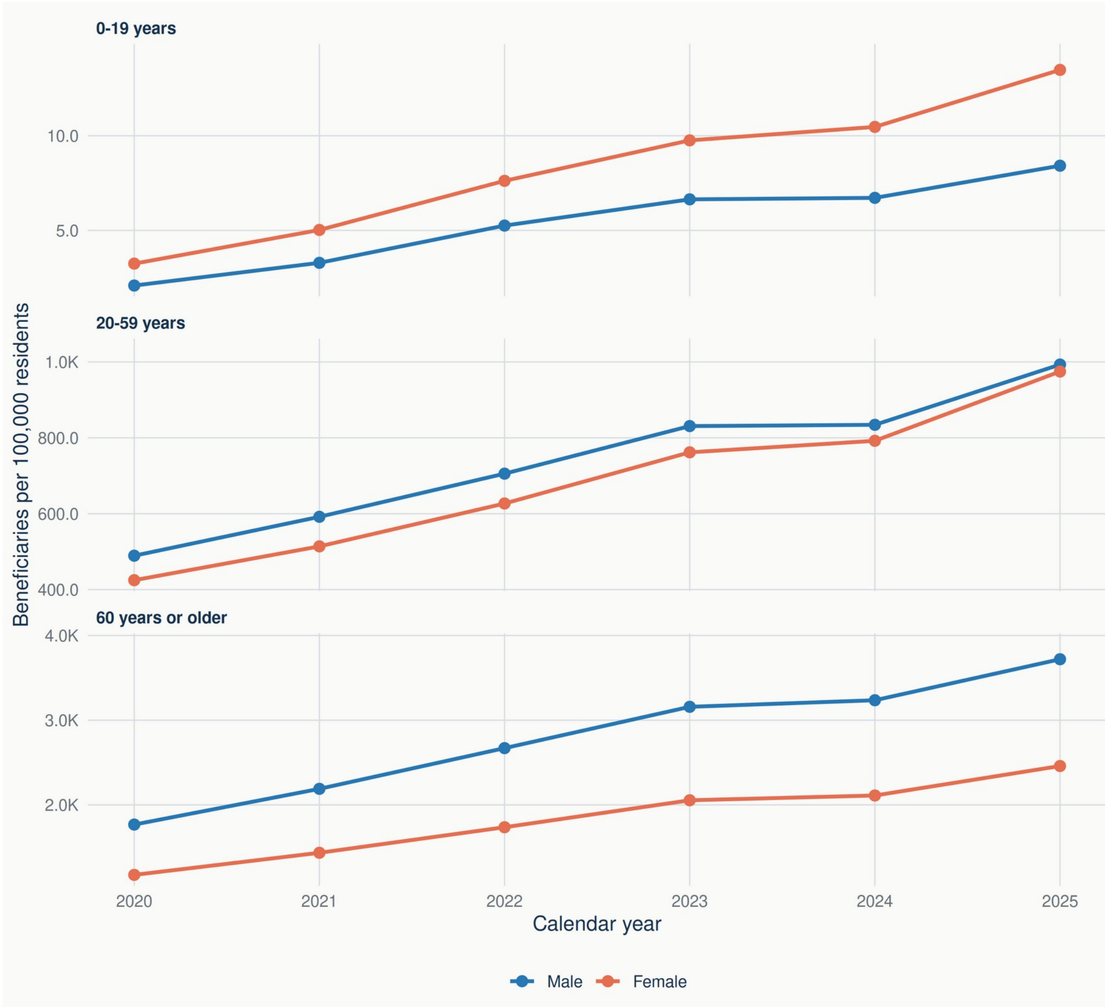

Figure 3. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in France, 2020–2025. Age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Regional variation

In 2025, metropolitan standardised rates varied twofold (Table 4; Figures 4 and 5). Hauts-de-France had the highest metropolitan rate (1,747.2 per 100,000), followed by Grand Est (1,568.2) and Bourgogne-Franche-Comté (1,464.3). Bretagne had the lowest rate (868.2), followed by Pays de la Loire (1,093.1) and Auvergne-Rhône-Alpes (1,160.7). The metropolitan maximum-to-minimum ratio was 2.01 and the coefficient of variation was 17.4%. Age-sex standardisation changed some rankings—notably increasing Île-de-France relative to its crude rate—but did not remove geographic heterogeneity.

The combined overseas grouping had 48,053 beneficiaries, a crude rate of 2,106.4 and a standardised rate of 2,435.9 per 100,000. It is reported separately from the metropolitan ranking because Open Medic does not provide the five overseas regions and departments individually. The figure map therefore does not colour or outline individual overseas territories.

Table 4. Regional beneficiary counts and age- and sex-standardised rates for reimbursed GLP-1 receptor agonist dispensings, France, 2025.

Regional analytical grouping	Beneficiaries	Standardised rate per 100,000	Change from 2024	Reimbursed expenditure
Overseas regions/departments	48,053	2,435.9	9.1%	€42.3m
Hauts-de-France	100,018	1,747.2	14.4%	€74.5m
Grand Est	88,986	1,568.2	18.2%	€65.1m
Bourgogne-Franche-Comté	43,640	1,464.3	16.8%	€32.2m
Centre-Val de Loire	37,575	1,395.4	19.0%	€26.8m
Provence-Alpes-Côte d'Azur/Corse	83,272	1,380.0	18.6%	€57.2m
Île-de-France	149,392	1,323.2	16.0%	€97.3m
Normandie	44,999	1,296.9	18.5%	€33.5m
Nouvelle-Aquitaine	83,386	1,229.9	20.6%	€62.2m
Occitanie	79,663	1,209.1	20.0%	€57.1m
Auvergne-Rhône-Alpes	95,294	1,160.7	20.2%	€69.1m
Pays de la Loire	43,553	1,093.1	18.5%	€33.0m
Bretagne	32,023	868.2	23.3%	€23.8m

Note: The overseas row is a combined grouping and is not comparable to a single administrative region. Provence-Alpes-Côte d'Azur and Corse form one analytical grouping. Regional beneficiary counts are independently deduplicated and are not additive: their 2025 sum (929,854) exceeds the independently deduplicated national count (925,644) by 4,210 (0.45%). Boxes and expenditure are additive. Expenditure is nominal. Sources: Open Medic; INSEE.

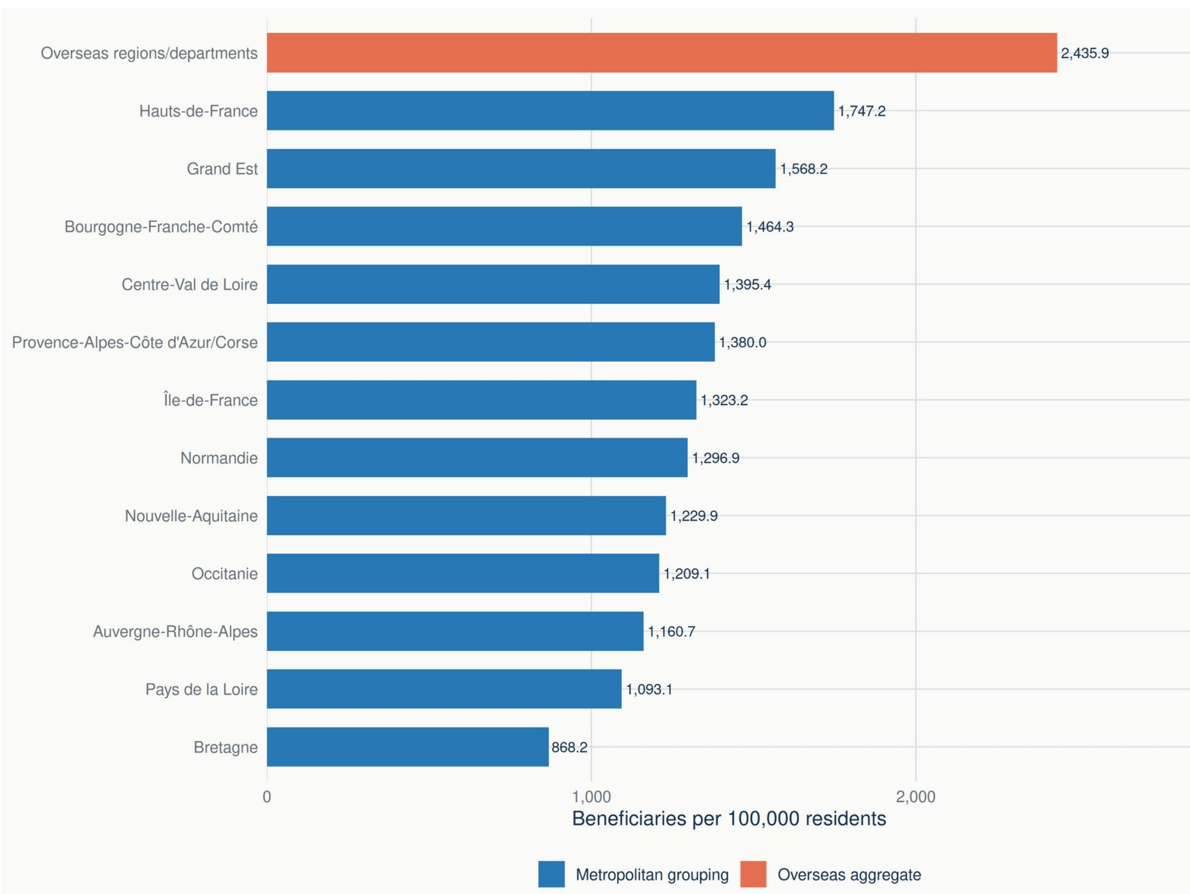

Figure 4. Age- and sex-standardised rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents across regional analytical groupings in France, 2025, ranked by rate. Provence-Alpes-Côte d'Azur and Corsica form a single analytical grouping; overseas regions and departments are available only as a combined grouping. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

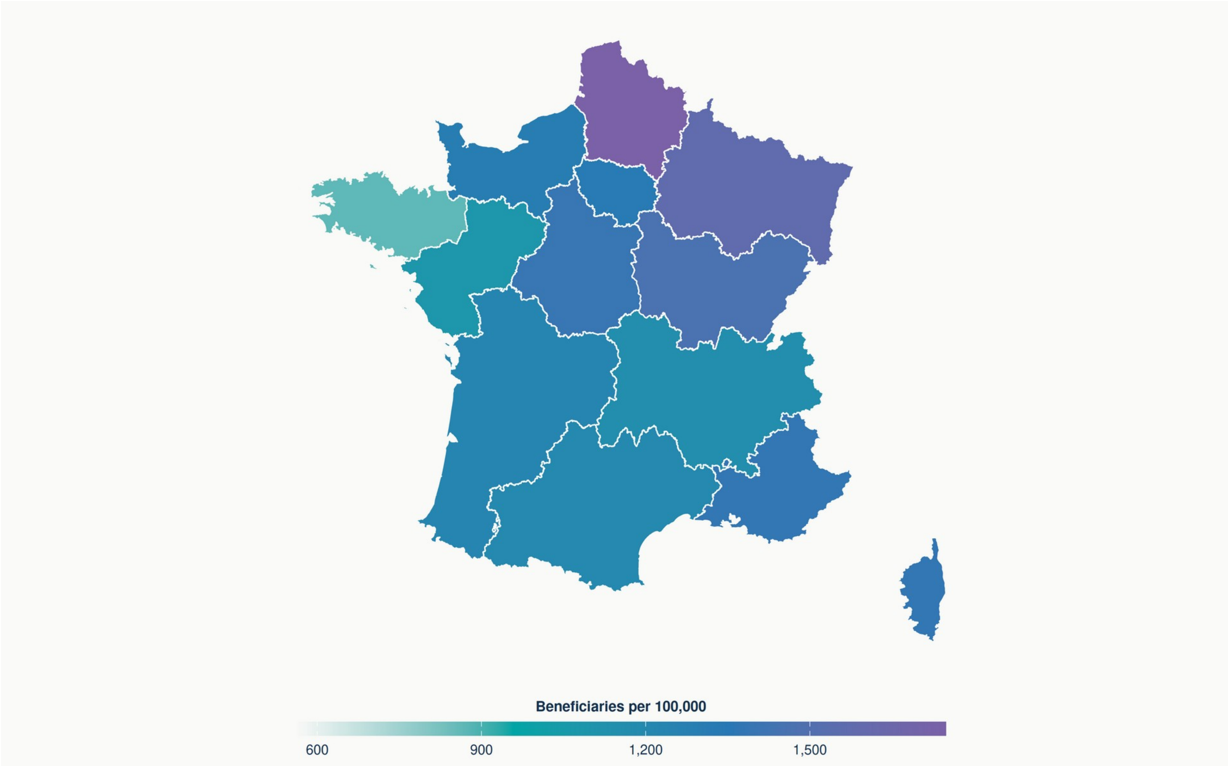

Figure 5. Age- and sex-standardised rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents across metropolitan regional groupings, 2025. Provence-Alpes-Côte d'Azur and Corsica form a single analytical grouping and therefore share the same estimate. Overseas territories are available only as a combined grouping (2,435.9 per 100,000) and are not mapped separately. Sources: Open Medic; INSEE population estimates; Etalab administrative boundaries derived from IGN ADMIN EXPRESS.

Regional evolution and supplementary profiles

The 2024 slowdown was geographically broad but not uniform. Metropolitan year-on-year changes in standardised rates ranged from −1.0% in Bourgogne-Franche-Comté to +5.3% in Provence-Alpes-Côte d'Azur/Corse; the median increase was 1.0%. In 2025, every metropolitan grouping increased, with gains ranging from 14.4% in Hauts-de-France to 23.3% in Bretagne (median 18.6%). The common-scale maps show that the north-eastern concentration of higher rates persisted while the overall distribution shifted upward.

The 2022–2025 common-scale maps are provided in Supplementary Figure S5. Complete age- and sex-specific regional profiles are provided for Île-de-France (S4-11), Centre-Val de Loire (S4-24), Bourgogne-Franche-Comté (S4-27), Normandie (S4-28), Hauts-de-France (S4-32), Grand Est (S4-44), the combined overseas grouping (S4-5), Pays de la Loire (S4-52), Bretagne (S4-53), Nouvelle-Aquitaine (S4-75), Occitanie (S4-76), Auvergne-Rhône-Alpes (S4-84), and Provence-Alpes-Côte d'Azur/Corse (S4-93).

Discussion

Principal findings

Reimbursed GLP-1 RA use in France expanded rapidly between 2020 and 2025: both annual beneficiaries and population rates more than doubled. Expansion was accompanied by a transformation of the substance mix, with semaglutide becoming the largest contributor to reimbursed boxes and expenditure by 2025. Use increased across all age-sex groups but remained concentrated among older adults, especially men. Large regional differences persisted after standardisation, indicating that population age and sex structure alone do not explain the geographic pattern.

Relation to previous evidence

Our findings extend individual-level French evidence showing growth in incident GLP-1 RA use through June 2023 [5]. The present analysis captures a broader annual beneficiary population and cannot classify indication, incident use or potential misuse, but it demonstrates that growth continued at the class level through 2025. The shift toward semaglutide is consistent with reports from Scandinavia and other settings [3,4]. Differences between studies in reimbursement systems, indications, products and units of measurement preclude direct rate comparisons, yet the direction of change is strikingly consistent.

The demographic pattern should be interpreted as utilisation, not comparative clinical response or need. Higher rates among older adults are compatible with the age distribution of type 2 diabetes and established reimbursement pathways during the study period. The near convergence of male and female rates among people aged 20–59 years by 2025, despite different baselines, may reflect changing indication mix, prescribing and access; aggregated data cannot identify which mechanism predominates.

The 2024 slowdown in context

Growth slowed temporarily in 2024, with only a marginal increase in reimbursed boxes and a small decline in nominal reimbursed expenditure, before accelerating again in 2025. This pattern coincided with documented supply constraints affecting several GLP-1 receptor agonists [9,10,15,16], although the aggregated data used in this study do not allow the slowdown to be causally attributed to product availability. The 2024 change should therefore be interpreted as a complementary observation within the broader multi-year expansion of reimbursed GLP-1 receptor agonist use.

Substance-level decomposition adds context without resolving causation: continued growth in semaglutide in 2024 was offset by contractions in liraglutide and dulaglutide. Open Medic cannot distinguish constraints on initiation from switching, discontinuation, stock management, changing indications, prescriber behaviour or demand. Monthly product-level analyses and individual treatment histories would be needed to separate these explanations, linking the annual pattern to the broader need for continued surveillance.

Overseas France

The combined overseas grouping produced the highest 2025 standardised rate. Because an unexpectedly high aggregate can arise from coding, denominator or version errors, we treated it as a dedicated quality-control question. The audit found exact agreement between direct and reconstructed denominators, reconciled direct and stratified numerators, reproduced crude and standardised rates, and verified corrected source files. Mayotte accounted for 14.5% of the combined 2025 denominator; eliminating the observed excess relative to the metropolitan median through denominator error alone would require a denominator 57.1% larger. These findings provide no evidence that the result is a simple computational artefact.

The estimate nevertheless has important limitations. Open Medic reports overseas regions and departments as one unit, so the rate cannot be assigned to Guadeloupe, Martinique, Guyane, La

Réunion or Mayotte individually. The territories differ substantially in demography, diabetes prevalence, healthcare access and population-estimate uncertainty. In addition, the numerator counts health-insurance beneficiaries with reimbursed dispensings while the denominator counts residents; mobility and coverage differences may be more consequential in overseas settings. The high combined rate is therefore a signal for further investigation, not evidence that every overseas territory has unusually high use. This is why individual overseas geometries were deliberately omitted from the map.

The age–sex-standardised rate for the combined overseas grouping was 15.6% higher than its crude rate (2,435.9 versus 2,106.4 per 100,000). This pattern is consistent with its younger age structure relative to the 2025 French standard population, which gives greater weight to older strata with higher observed utilisation rates. Importantly, the elevated standardised rate was not explained by population composition alone, as utilisation also remained comparatively high within several adult age–sex strata. This secondary observation reinforces the need for territory-specific investigation rather than territorial inference from the combined estimate.

Economic and policy implications

Nominal reimbursed expenditure for the studied class rose from €298.7 million in 2019 to €674.6 million in 2025. This measure is not a budget-impact analysis: it does not adjust for inflation, rebates, confidential price agreements, avoided healthcare costs, indication or outcomes. It nevertheless documents a rapidly expanding reimbursed pharmaceutical commitment. The concentration of spending in semaglutide and dulaglutide also makes expenditure sensitive to product-specific prices, supply and prescribing rules.

The prospective economic stakes are larger than the historical trend alone. HAS highlighted substantial uncertainty around the eligible population and estimated that reimbursing semaglutide for severe obesity could generate a multi-billion-euro three-year impact under broader population assumptions [7]. After the study period, France introduced restricted reimbursement for Wegovy and Mounjaro from June 2026 [17]. Because the present primary definition excludes tirzepatide and reimbursed obesity use was limited during 2019–2025, the study should be considered a pre-expansion baseline. Future surveillance must separate diabetes and obesity indications where possible, include dual agonists in a prespecified broader incretin analysis, and assess whether access expands equitably across demographic and geographic groups.

Strengths and limitations

Strengths include national coverage of reimbursed community dispensings, extension through 2025, integration of counts, rates, substance composition, expenditure and geography, prospective protocol governance, explicit amendments, source checksums, independently reproducible figure data and dedicated quality control of unusual results. Direct age-sex standardisation and a metropolitan sensitivity perspective reduce, but do not eliminate, compositional confounding.

Several limitations are inherent to aggregated reimbursement data. First, annual beneficiaries cannot be linked over time, so initiation, persistence, switching and adherence are not estimable. Second, indication and diagnosis are unavailable; reimbursed use cannot be separated into diabetes, obesity or other clinical contexts, and non-reimbursed purchases are absent. Third, boxes are not equivalent treatment quantities across products. Fourth, substance-specific beneficiary counts overlap. Fifth, expenditure is nominal and may differ from net payer cost. Sixth, Open Medic disclosure control created interval-valued standardised estimates for some regions in 2020–2021. Seventh, the analytical geography combines Provence-Alpes-Côte d'Azur with Corse and all overseas regions and departments into a single unit. Finally, aggregate regional associations cannot be interpreted at individual level or causally.

Implications for surveillance and research

- Update the open analysis annually and add monthly or quarterly sources where available to distinguish supply shocks from longer-term diffusion.
- Prespecify a broader incretin-therapy analysis that includes tirzepatide without changing the historical A10BJ series.
- Use individual-level SNDS analyses to estimate indication, initiation, persistence, switching, discontinuation and patient-level determinants.
- Seek territory-specific overseas data and evaluate numerator–denominator alignment before making territorial comparisons.
- Link utilisation surveillance to transparent economic evaluation, including net prices, treatment duration, displaced costs and health outcomes.

Conclusions

Reimbursed GLP-1 receptor agonist use expanded substantially in France through 2025, with annual beneficiaries more than doubling over five years and semaglutide becoming the dominant substance by box volume. A temporary slowdown in 2024 formed one part of this longer-term expansion and should not be interpreted causally from annual aggregate data. Persistent age-sex and regional differences—including a high but non-disaggregable overseas estimate—show the value of combining national trends with structured demographic, geographic and quality-control analyses. Open Medic can support timely, transparent surveillance, but individual-level data remain necessary to determine indication, clinical appropriateness, treatment trajectories and causal drivers. Given the scale of reimbursed expenditure and the subsequent expansion of obesity-treatment coverage, continued monitoring is both a public-health and economic priority.

Declarations

Data availability

Open Medic and INSEE population data are publicly available from the sources cited below. The public repository accompanying this study is maintained at https://github.com/javierzsm/glp1-use-france and contains version-controlled code, metadata, quality-control outputs and figure-linked CSV files intended to permit reconstruction of every reported figure. An interactive exploration of the results is available at https://javierzsm.github.io/glp1-use-france/.

Ethics

Only aggregated, open and disclosure-controlled data were analysed. No identifiable individual-level data were accessed; ethics approval and individual consent were not required.

Funding and competing interests

Funding: This study was internally funded by Maradian Labs. No external commercial, institutional or grant funding was received. Competing interests: The author declares no competing interests.

Author contributions

Javier Zorrilla de San Martin: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, and Writing – review and editing.

Use of artificial intelligence

OpenAI ChatGPT and Codex were used to assist with manuscript drafting and language editing, code development, figure and dashboard production, and quality-control workflows. No identifiable individual-level health data were provided to these tools. The author made all scientific and methodological decisions, reviewed and verified the AI-assisted outputs, and assumes full responsibility for the accuracy and integrity of the work. Artificial intelligence tools were not credited as authors.

About the author

Javier Zorrilla de San Martin develops open, reproducible analytical projects at the intersection of biomedical research, drug utilisation and real-world health data. His work combines research design, statistical analysis, reproducible computational workflows and transparent scientific communication. This study forms part of his public research and data-analysis portfolio.

Suggested citation

Zorrilla de San Martin J. Reimbursed use of GLP-1 receptor agonists in France, 2019–2025: national, active-substance, demographic and regional trends from Open Medic data. Research report; 2026. DOI: https://doi.org/10.5281/zenodo.22658978. Repository: https://github.com/javierzsm/glp1-use-france.

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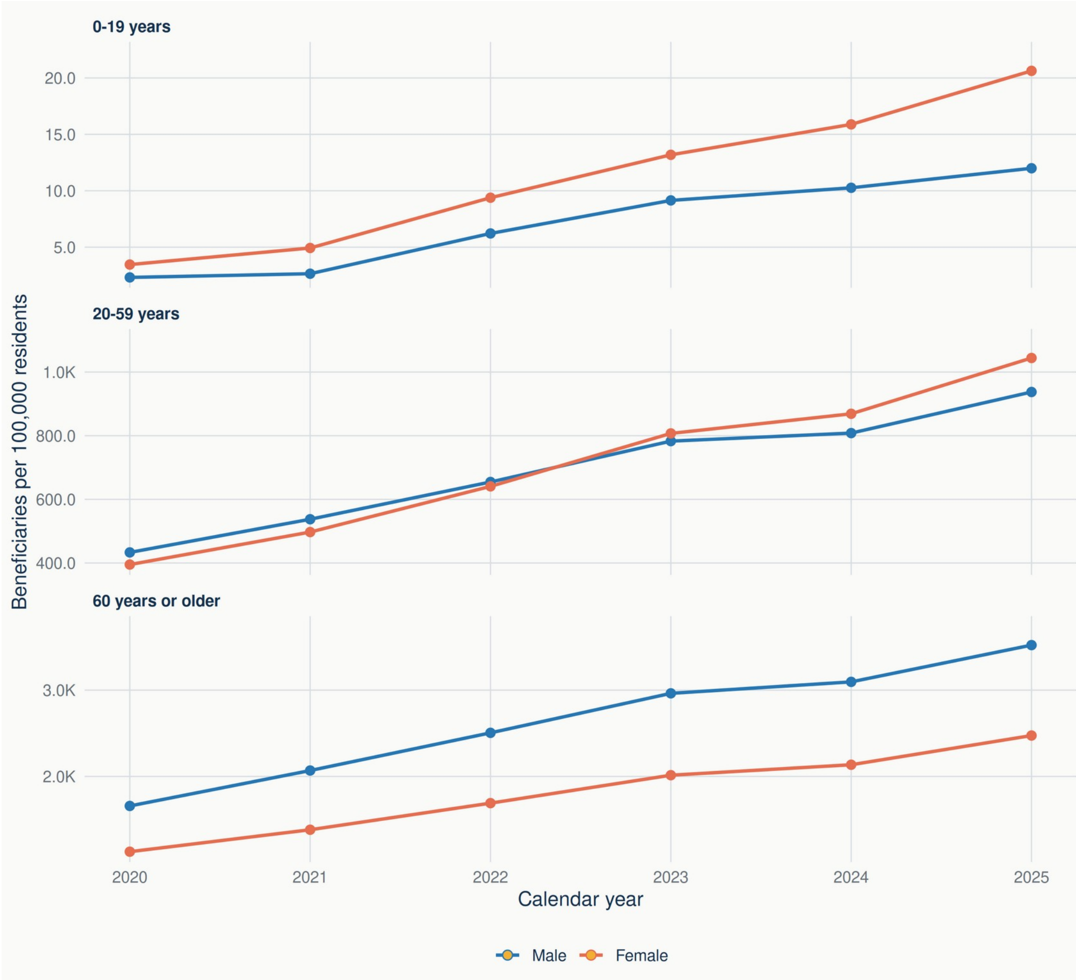
Supplementary figure appendix

This appendix contains all prespecified supplementary figures. Each figure is linked from the Results section. Regional age-sex panels use independent vertical scales by age group; vertical ranges represent uncertainty caused by disclosure control rather than sampling confidence intervals. Gold-filled markers denote supported structural zeros.

Appendix navigation

Supplementary Figure S4-11



Supplementary Figure S4-11. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Île-de-France, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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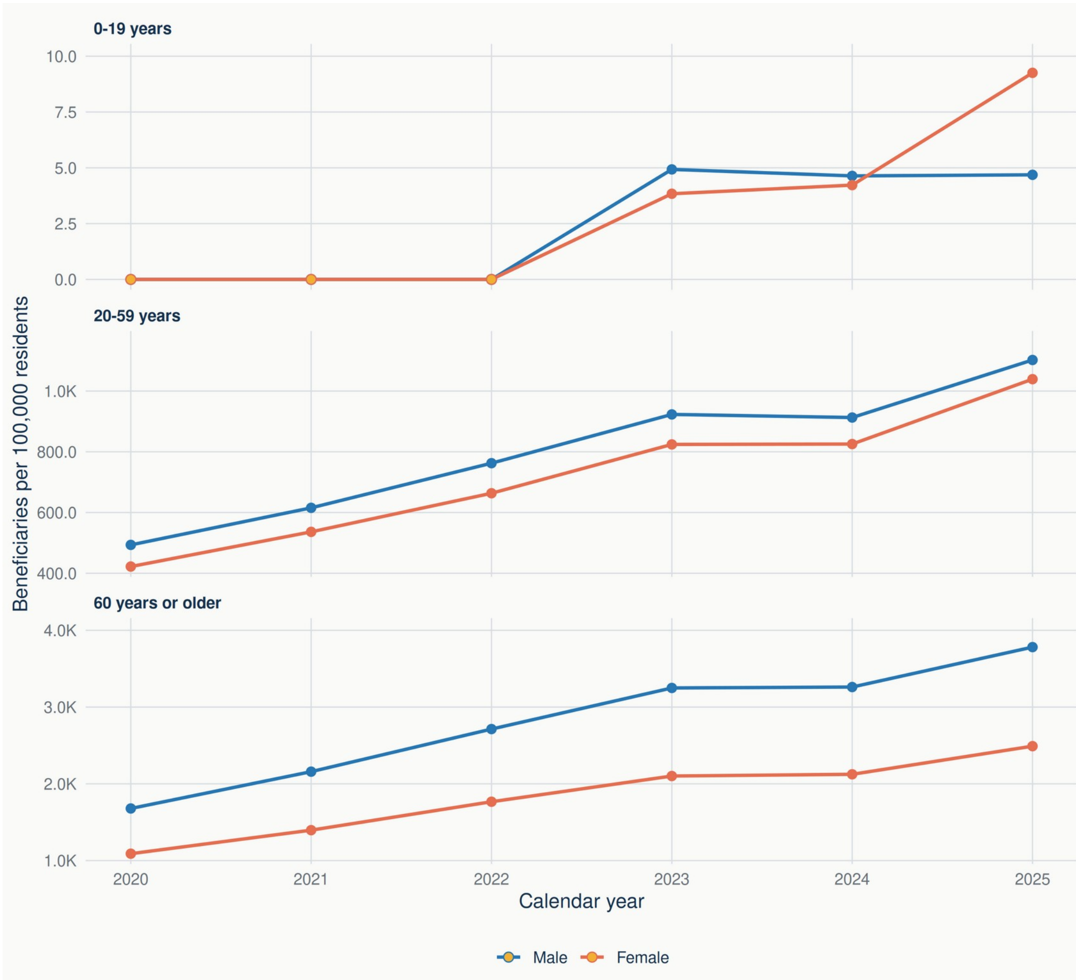
Supplementary Figure S4-24



Supplementary Figure S4-24. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Centre-Val de Loire, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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Supplementary Figure S4-27

Supplementary Figure S4-27. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Bourgogne-Franche-Comté, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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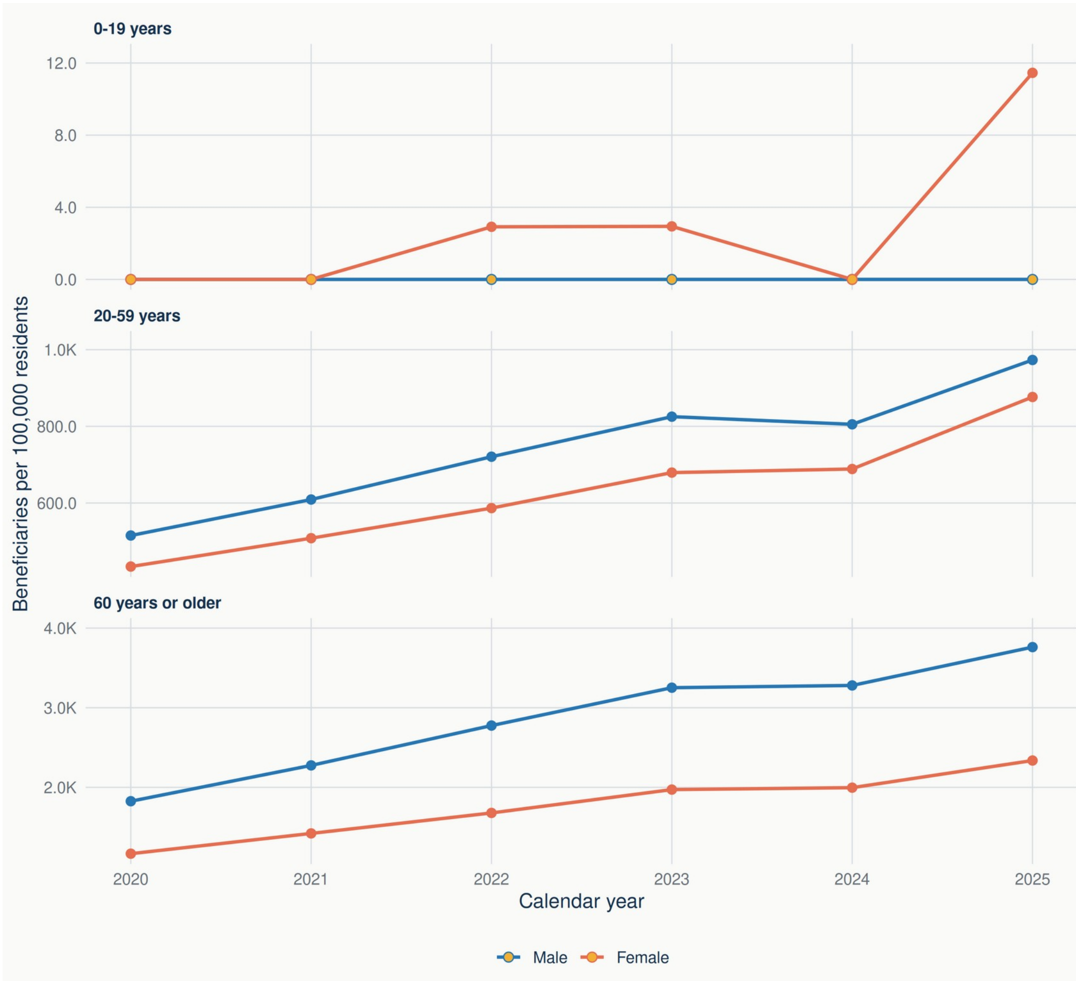
Supplementary Figure S4-28



Supplementary Figure S4-28. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Normandie, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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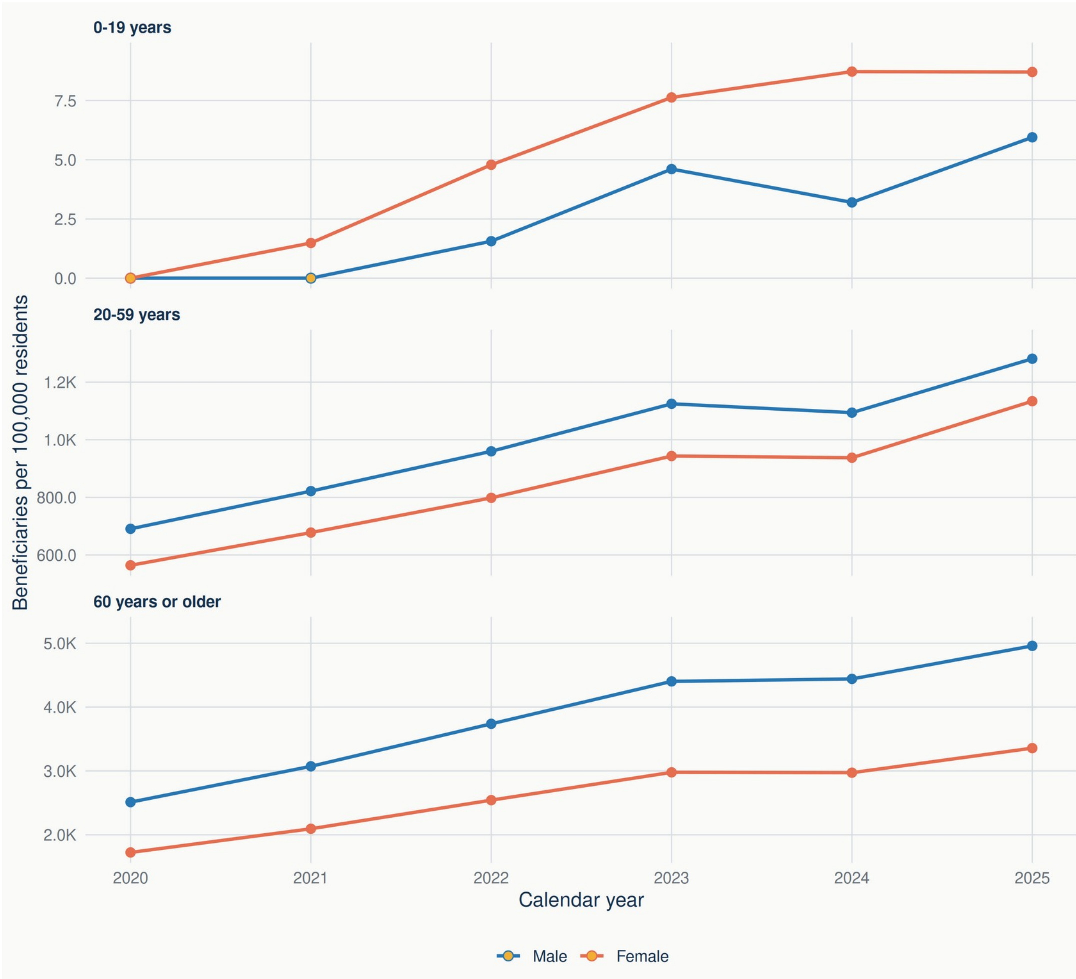
Supplementary Figure S4-32



Supplementary Figure S4-32. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Hauts-de-France, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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Supplementary Figure S4-44

Supplementary Figure S4-44. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Grand Est, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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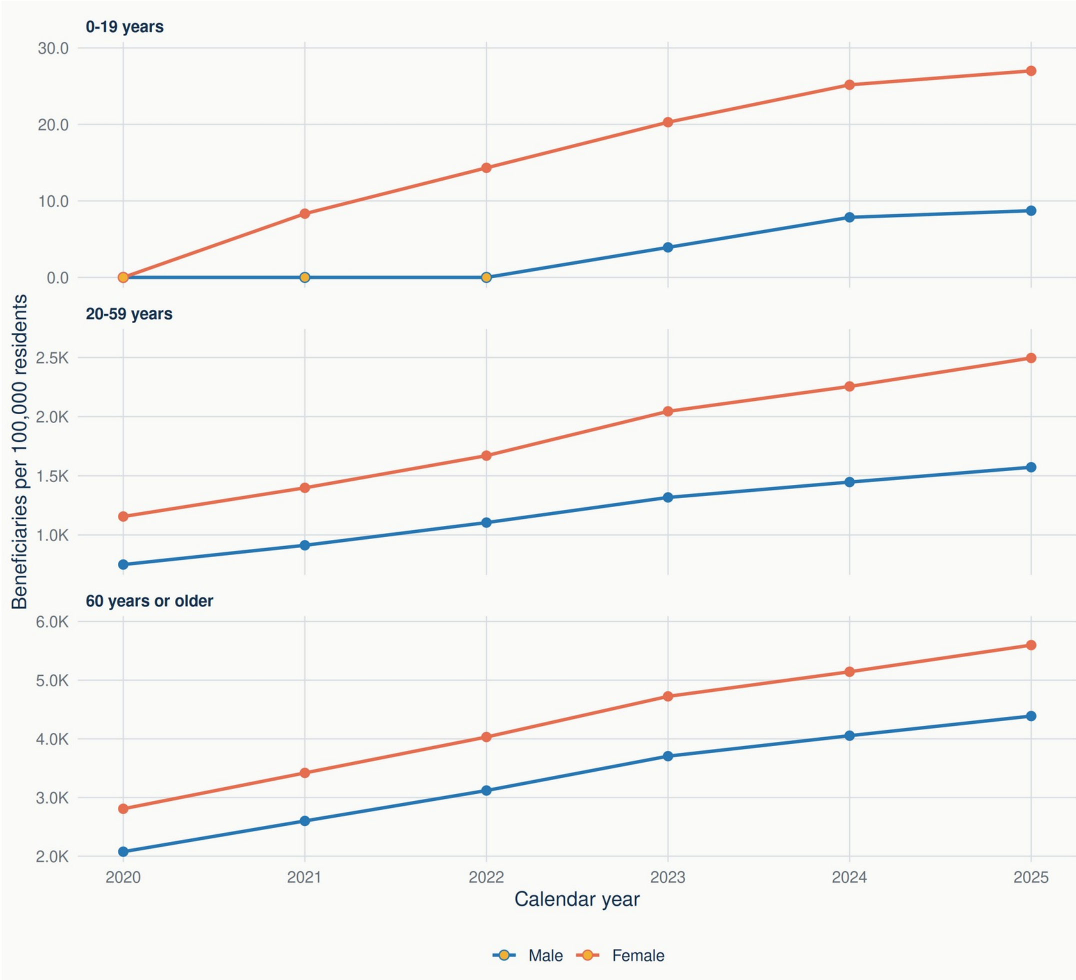
Supplementary Figure S4-5



Supplementary Figure S4-5. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Overseas regions and departments, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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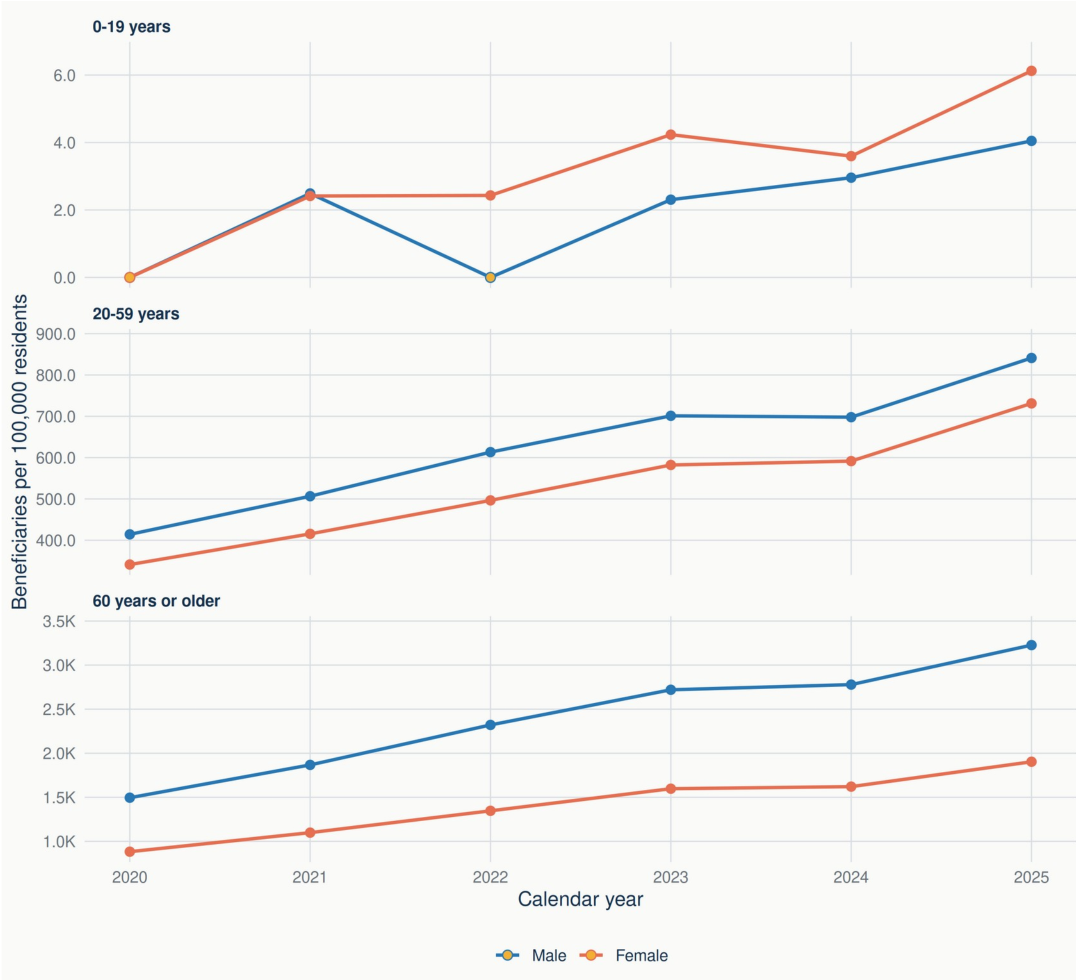
Supplementary Figure S4-52



Supplementary Figure S4-52. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Pays de la Loire, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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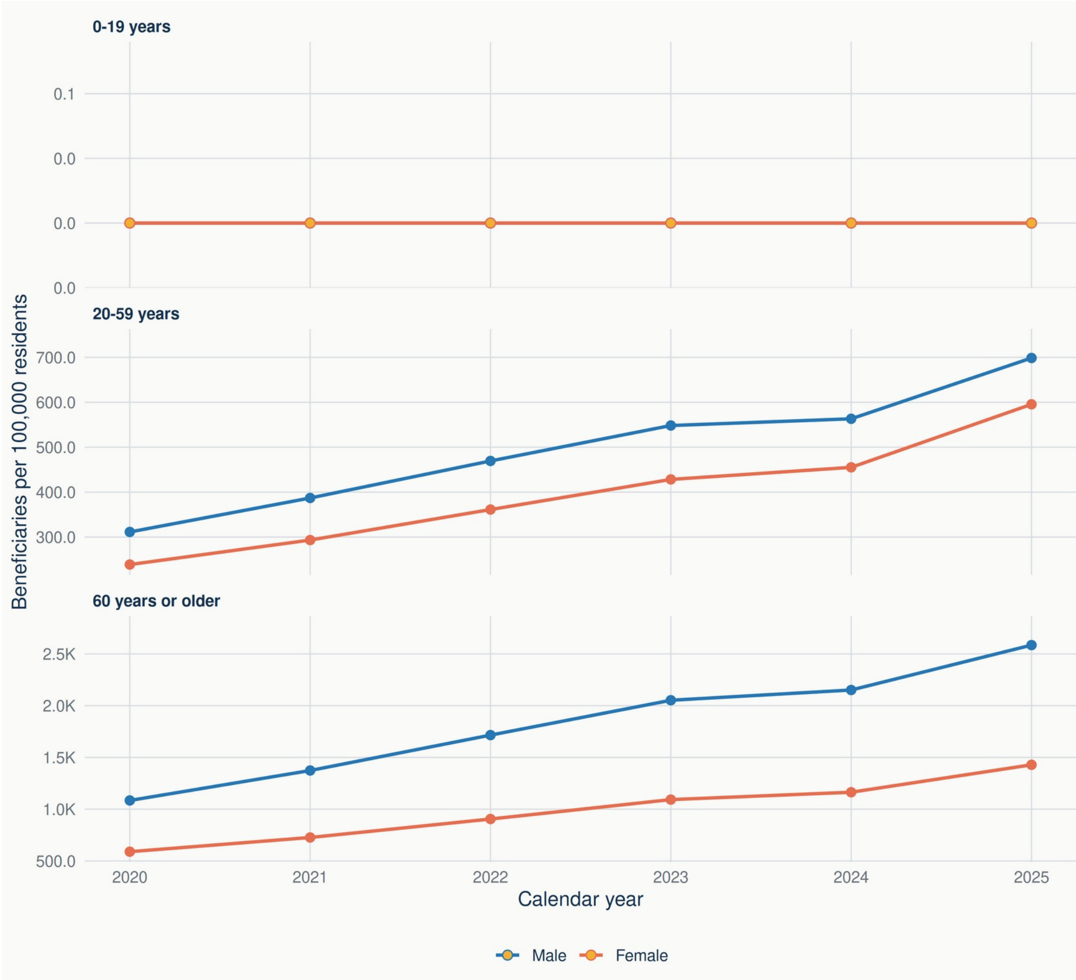
Supplementary Figure S4-53



Supplementary Figure S4-53. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Bretagne, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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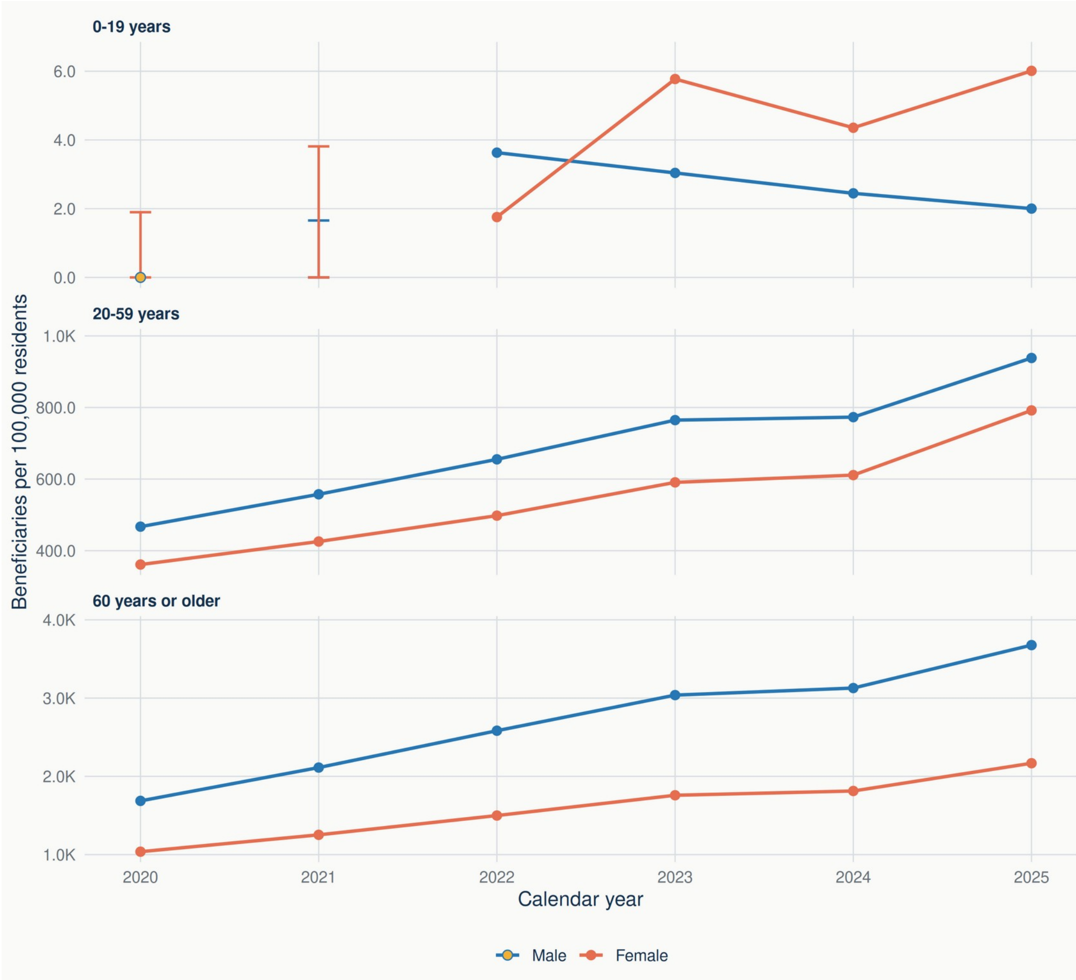
Supplementary Figure S4-75



Supplementary Figure S4-75. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Nouvelle-Aquitaine, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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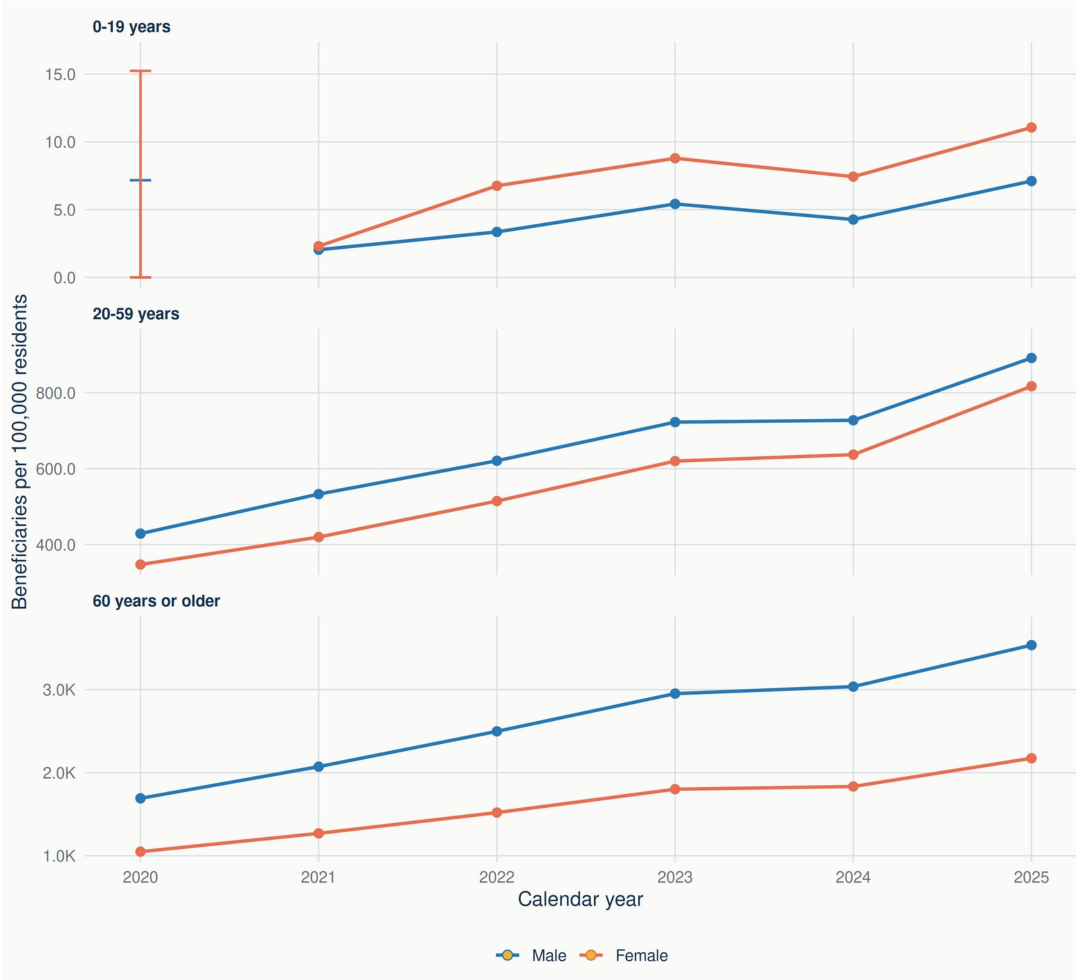
Supplementary Figure S4-76



Supplementary Figure S4-76. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Occitanie, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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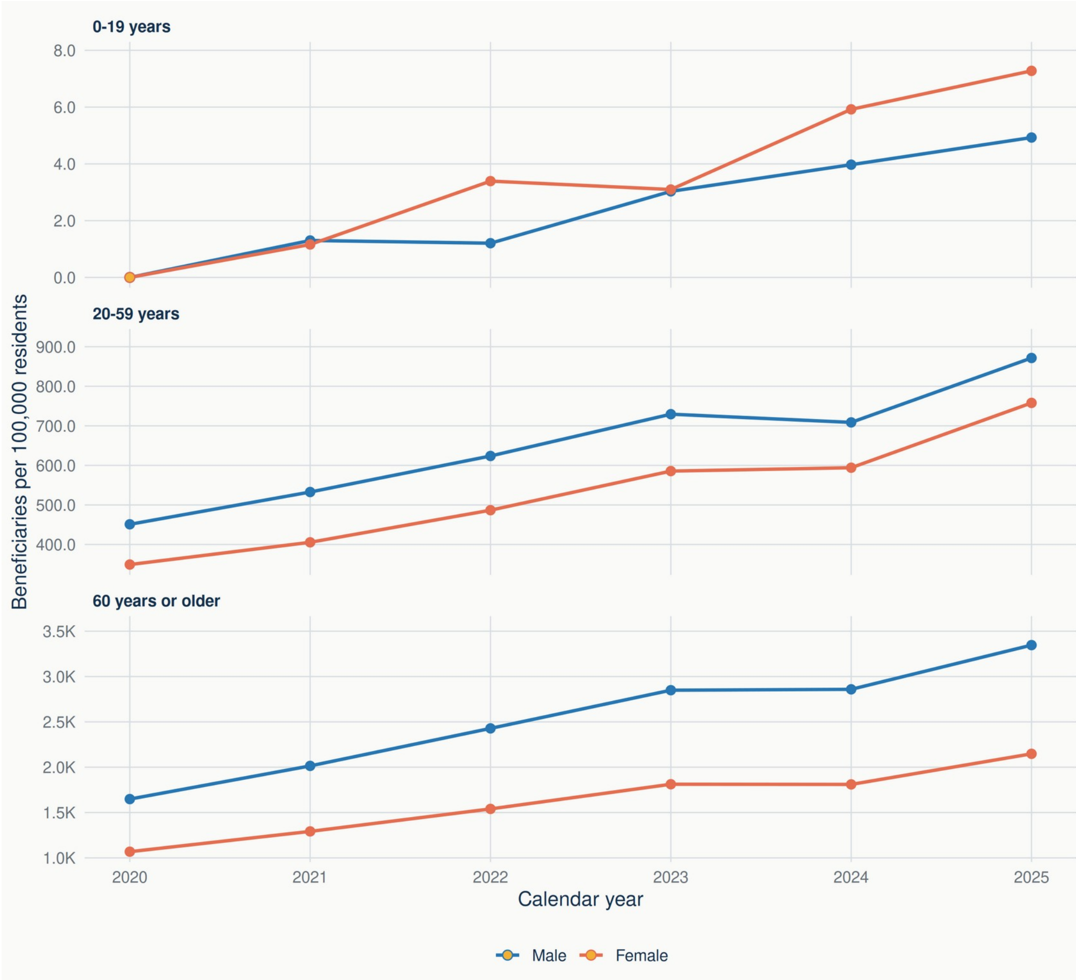
Supplementary Figure S4-84



Supplementary Figure S4-84. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Auvergne-Rhône-Alpes, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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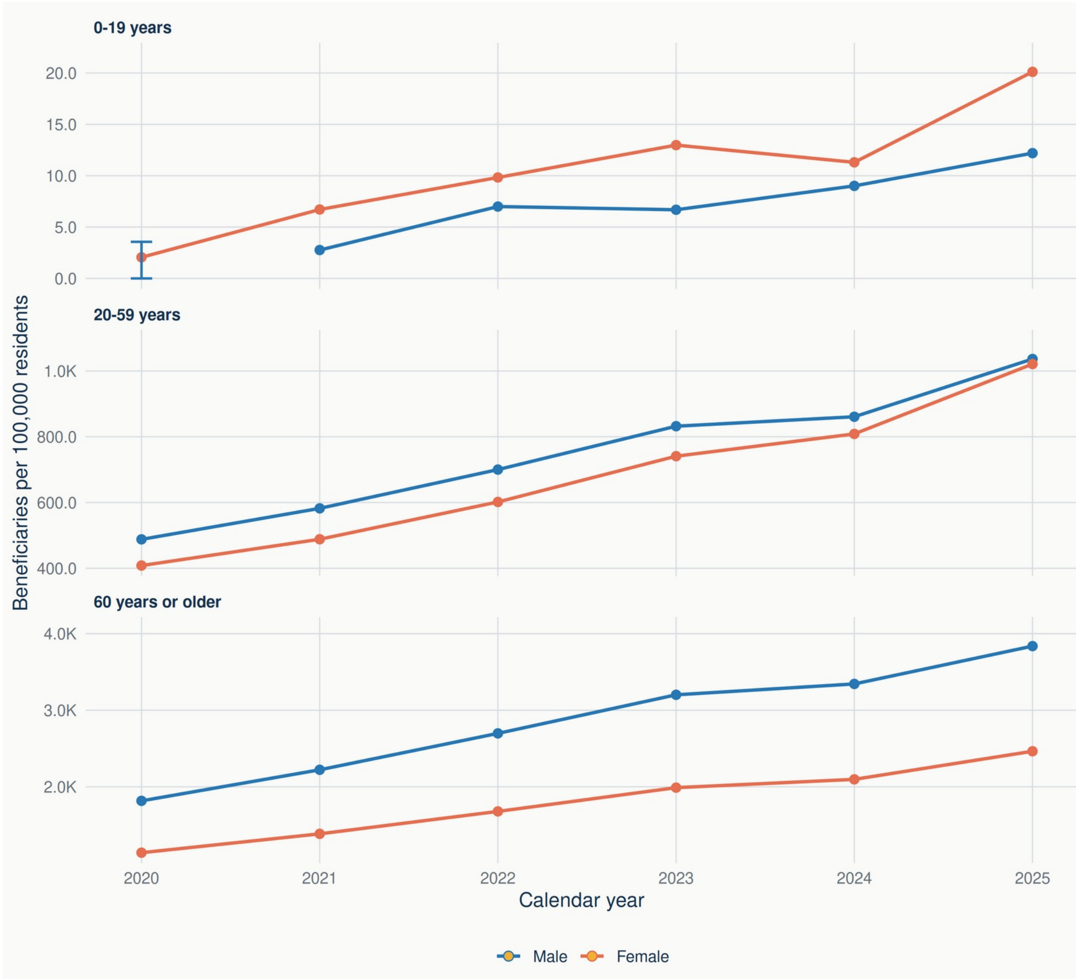
Supplementary Figure S4-93



Supplementary Figure S4-93. Age- and sex-specific rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents in Provence-Alpes-Côte d'Azur and Corse, 2020–2025. Vertical ranges represent disclosure-control bounds, not confidence intervals; gold-filled markers denote supported structural zeros; age-group panels use independent vertical scales. Sources: Open Medic; French National Institute of Statistics and Economic Studies (INSEE).

Note: Vertical ranges shown for affected cells in 2020 and, where applicable, 2021 are disclosure-control bounds. They allocate between zero and all compatible beneficiaries with unknown age to the displayed stratum and represent data-availability uncertainty, not sampling error or confidence intervals. Sources: Open Medic; INSEE.

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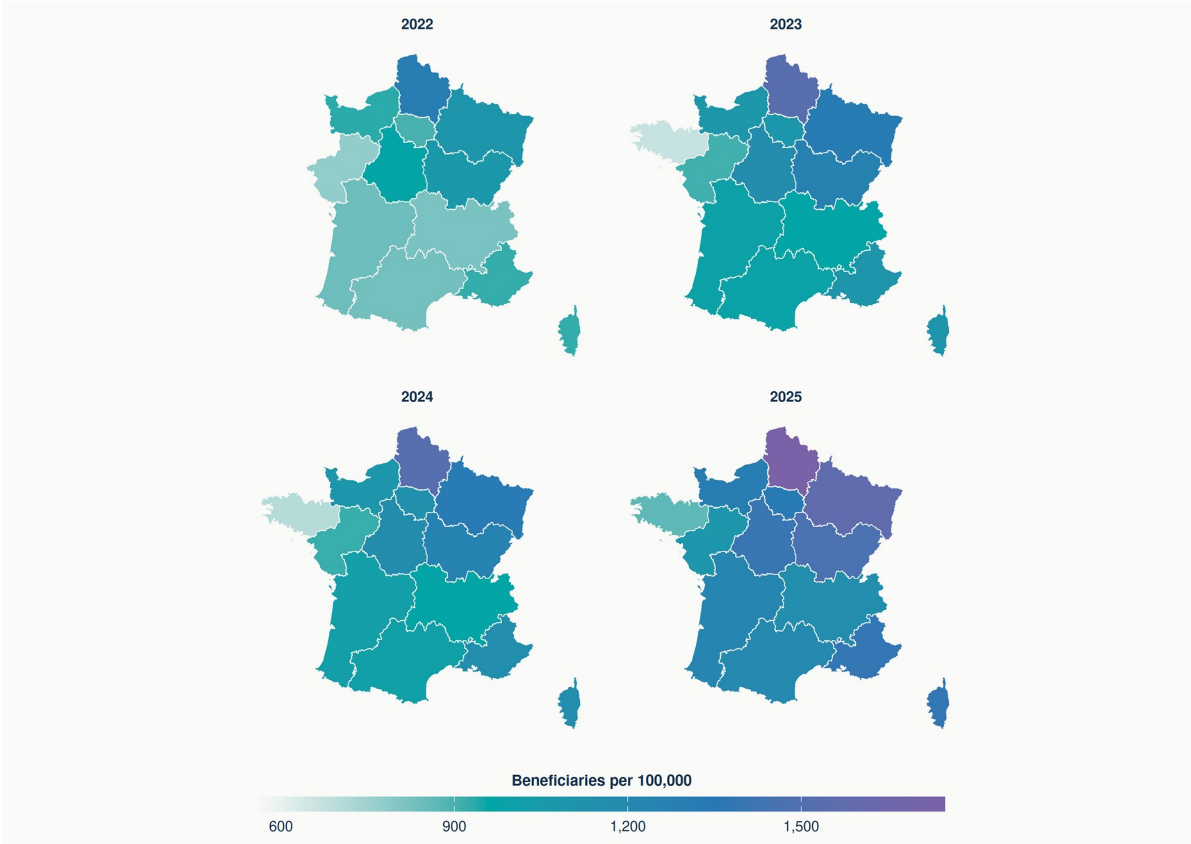
Supplementary Figure S5



Supplementary Figure S5. Age- and sex-standardised rates of beneficiaries with reimbursed GLP-1 receptor agonist dispensings per 100,000 residents across metropolitan regional groupings, 2022–2025. All panels use the same colour scale. Results for 2020–2021 are not mapped because disclosure control produced intervals rather than complete point estimates. Provence-Alpes-Côte d'Azur and Corsica form a single analytical grouping. Overseas territories are available only as a combined grouping and are not mapped separately. Sources: Open Medic; INSEE population estimates; Etalab administrative boundaries derived from IGN ADMIN EXPRESS.

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RECORD-PE reporting checklist

This checklist maps the STROBE, RECORD and RECORD-PE reporting domains to the manuscript. Locations identify named sections, tables or figures rather than mutable page numbers. Items that require individual-level records, database linkage, exposure episodes, validation algorithms or comparative causal analyses are marked not applicable and explained. The checklist is based on Langan et al. [18].

Item	Reporting requirement	Location in manuscript	Applicability and notes
1; RECORD 1.1–1.3	Identify the design, routinely collected data, geography, timeframe and linkage in the title or abstract.	Title; Executive summary—Methods	Met. Open Medic, France and 2019–2025 are explicit; no record linkage was performed.
2	Explain scientific background and rationale.	Introduction	Met.
3	State specific objectives and prespecified hypotheses.	Introduction, final paragraph; Methods—Study design and governance	Met. Objectives were descriptive; no null hypotheses were planned.
4; PE 4.a–4.b	Present study design and relevant design features early; use a design diagram when useful.	Methods—Study design and governance; Methods—Historical ATC harmonisation	Met. Repeated annual cross-sectional design. A participant-flow diagram is not informative for aggregate annual cells.
5	Describe setting, locations and relevant dates.	Methods—Data sources; Methods—Historical ATC harmonisation	Met.
6; RECORD 6.1–6.3; PE 6.1.a	Define eligibility, selection, codes/algorithms and any matching.	Methods—Study population and pharmacological definition	Met. ATC definitions and eligible aggregates are reported. Matching and validation of a patient-identification algorithm are not applicable.
7; RECORD 7.1; PE 7.1.a–g	Define outcomes, exposures, confounders, modifiers and code lists; describe exposure construction and validation.	Methods—Study population and pharmacological definition; Methods—Measures and statistical analysis	Partly applicable. Measures and ATC codes are defined. Exposure episodes, dose, adherence, validation and confounding variables are unavailable in aggregated data.
8; PE 8.a	Describe data sources, measurement and how the healthcare system generates records.	Methods—Data sources; Methods—Study population and pharmacological definition	Met. The reimbursement setting and community-pharmacy scope are stated.
9	Describe efforts to address bias.	Methods—Historical ATC harmonisation; Methods—Overseas grouping audit; Discussion—Strengths and limitations	Met.
10	Explain study size.	Methods—Study design and governance; Methods—Data sources	Met. All eligible aggregate records were used; no sample-size calculation was applicable.
11	Explain handling of quantitative variables and groupings.	Methods—Measures and statistical analysis	Met.
12; RECORD 12.1–12.3; PE	Describe statistical methods, data cleaning, linkage, temporal ordering and sensitivity analyses.	Methods—Measures and statistical analysis; Methods—Overseas grouping audit; Methods	Met where applicable. There was no individual-level linkage, time-varying exposure model

Item	Reporting requirement	Location in manuscript	Applicability and notes
12.1.a–b		—Reproducibility, reporting and ethics	or confounder adjustment.
13	Report observations at each stage and reasons for exclusion.	Methods—Historical ATC harmonisation; Results; Tables 1–4	Adapted to aggregate data. The 2019 class-level beneficiary measure excluded after QC is explained.
14	Provide descriptive characteristics, missingness and follow-up information.	Results—National evolution; Age and sex; Regional variation; Tables 1–4	Met where applicable. Individual follow-up and participant-level covariates are unavailable.
15	Report outcome events or summary measures.	Results; Tables 1–4; Figures 1–5	Met.
16	Report main results, precision and category boundaries.	Results; Tables 1–4; Figures 1–5	Met for descriptive estimates. Disclosure-control bounds are reported; sampling confidence intervals are not applicable.
17	Report other and sensitivity analyses.	Methods—Overseas grouping audit; Results—Regional variation; Discussion—Overseas France	Met.
18	Summarise key results with reference to objectives.	Discussion—Principal findings	Met.
19; RECORD 19.1; PE 19.1.a	Discuss limitations, data quality, misclassification and incomplete exposure capture.	Discussion—Strengths and limitations	Met, including absent indications, non-reimbursed use, disclosure control and geographic aggregation.
20; PE 20.a	Provide cautious interpretation considering multiplicity, bias and comparable evidence.	Discussion—Relation to previous evidence; The 2024 slowdown in context; Overseas France	Met. Confounding by indication and causal effect estimation are not applicable to the descriptive design.
21	Discuss generalisability.	Discussion—Strengths and limitations; Conclusions	Met.
22; RECORD 22.1	Report funding, conflicts and access to protocol, code and data.	Document status; Declarations—Data availability; Funding and competing interests	Met. Public repository URL is provided; source data are open.

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